FINAL PRODUCT PLAN

Centralised Admission Management System

Multi-Branch • Multi-Course • Multi-Cycle • Centralised Operations

Deployment model	One centralised application and one central database
Organisation	Multiple branches, each offering multiple courses
Minimum capacity	500,000 applicant master records, with expansion headroom
Application intake	External portal retained; application data and documents imported
Technology	Laravel, Vue.js, MySQL 8, Redis and private object storage

Scope principle

The external portal remains responsible for online application submission. This system imports applicant data and uploaded documents and manages admission processing from scrutiny/shortlisting onwards. It does not recreate online application submission or operate a payment gateway.

Document structure

1. Executive summary and objectives
2. Scope and workflow
3. Centralised organisational model
4. Roles and access control
5. Functional modules
6. Import and document integration
7. Selection, GD/PI, merit and allocation
8. Payments and refunds
9. Database schema
10. API catalogue
11. Technical architecture and capacity
12. Security and privacy
13. Reports
14. Development phases
15. Testing and acceptance
16. Decisions before implementation

1. Executive summary and objectives

The product is one centralised admission platform for a single institution operating multiple branches. Each branch may offer several courses. Admission committees, admission cells, accounts teams and faculty panels are configured and managed in the same platform and database. Central administration has institution-wide oversight, while users receive permissions scoped to their authorised branches, courses, cycles, rounds and assignments.

The baseline capacity is at least 500,000 applicant master records. The architecture should scale beyond this through indexed database design, background processing, scalable document storage, pagination and realistic load testing.

• Import applicant records and supporting documents from the existing external portal.

• Maintain a common applicant identity and separate applications for each course and cycle.

• Manage scrutiny, eligibility, GD/PI, merit, reservation, allocation, offers and confirmation.

• Support multiple branches, courses, committees, admission cells and accounts teams centrally.

• Record externally received payments and refunds without a payment gateway.

• Provide secure, auditable and consolidated reporting.

2. Scope and end-to-end workflow

Stage	System responsibility
External portal	Applicant submits online application and uploads documents in the existing portal.
Import management	Staff import data through Excel, CSV, API, ZIP or secure document URLs.
Validation	Map fields, validate records, identify duplicates/missing data, preview and approve.
Scrutiny and eligibility	Verify documents and assess course-specific eligibility.
Shortlisting	Apply approved shortlisting rules and prepare candidate lists.
GD/PI	Create panels, schedule candidates, assign faculty and record evaluations.
Merit	Calculate course-wise composite scores using approved, versioned criteria.
Allocation	Apply approved seat and reservation rules and allocate branch/course seats.
Offers and waitlists	Issue offers, record responses, manage vacancies and waitlist movement.
Admission and accounts	Record confirmation, manually verify payments and manage refund records.
Reporting and closure	Produce reports and lock completed rounds.

Explicit exclusions

• Online application form, applicant registration and application submission.

• Online application-fee collection or payment gateway integration.

• Automatic bank debit, payment initiation or automatic refund execution.

3. Centralised organisational model

Centralisation means one software installation, one logical database, one identity/access framework and a common reporting layer. Branches are organisational records inside the central system—not separate installations or isolated databases.

Level	Configuration and data
Institution	Institution profile, global settings, governance and consolidated dashboards.
Branch	Branch code, name, location, contacts, status and staff assignments.
Course	Course code/title, eligibility, selection policy and fee structure.
Branch-course offering	Defines which course is offered by which branch in a cycle.

Level	Configuration and data
Admission cycle	Academic year/cycle, dates, status and participating course offerings.
Admission round	Initial/fresh round, seat matrix, rule version, merit and approval history.
Committee/team	Constitution, members, roles, validity, scope and approval powers.

Illustrative hierarchy

Institution → Branches → Course offerings → Admission cycles → Admission rounds. A branch may offer multiple courses, and the same course may be offered by multiple branches. Course offerings may have different selection policies, committees, fee structures, GD/PI panels and seat matrices. Keep branch and physical campus/location as separate concepts.

4. Roles and access control

Role	Main responsibility
Super Admin	System setup, users, roles, permissions, integrations and global audit.
Central Admission Administrator	Cycles, offerings, import governance and central operations.
Branch Administrator	Authorised branch-level operations.
Committee Chairperson	Review and approve committee recommendations within delegated authority.
Committee Member	Participate in selection and committee decisions within assigned scope.
Admission Cell Officer	Scrutiny, eligibility, scheduling, offers and admission processing.
Admission Cell Staff	Document processing and assigned administrative tasks.
Accounts Officer	Review payment entries and approve/reject/return; oversee refunds.
Accounts Staff	Enter payment particulars, evidence and actual refund transactions.
Faculty / Panel Member	View assigned candidates and submit individual evaluations.
Reporting Officer	View approved reports within assigned scope.

- Enforce permissions on the server; menu hiding alone is not security.
- Scope access by branch, course, cycle, round and panel assignment.
- Allow multiple roles where authorised, with controls for conflicting duties.
- Record actor, timestamp, decision and remarks for approvals.
- Faculty can access only assigned candidates and evaluation tasks.

5. Functional modules

5.1 Master configuration

Institution, branches, locations, courses, branch-course mappings, academic years, cycles, rounds, categories, document types, fee heads, payment modes and settings.

5.2 Applicant/application management

Common applicant master; separate course/cycle application; external application reference; contact details, academic records, entrance data, category, preferences and status history.

5.3 Import management

Excel/CSV/API import, mapping, staging, validation, preview, duplicate detection, approval, execution, retry and error reports.

5.4 Document management

Secure storage, classification, versioning, verification, missing-document tracking, authorised downloads and retention.

5.5 Scrutiny and eligibility

Configurable checklists, course-specific criteria, verification decisions, deficiency notices and approval history.

5.6 Entrance and shortlisting

Import entrance scores, map and validate scores, configure thresholds/rules and publish approved shortlists.

5.7 GD/PI scheduling

Panels, chairpersons, members, slots, candidate assignments, venues/links and conflict prevention.

5.8 Evaluation

Configurable criteria and marks; individual scoring; remarks; submit/lock; controlled reopening; aggregation and outlier flags.

5.9 Merit calculation

Versioned course-specific formula, normalisation, calculation breakdown, tie-breaks, simulation and approval.

5.10 Reservation and allocation

Approved category rules, seat matrices, allocation simulation, committee approval and publication.

5.11 Offers and waitlists

Offer generation, response, acceptance, decline, expiry, vacancy tracking, waitlist movement and confirmation.

5.12 Fresh admission rounds

Separately authorised fresh rounds with their own seat matrix, policy version, merit and allocation history.

5.13 Payments and refunds

Manual payment entry, evidence, review, approval/rejection/return, refund request, approval, transaction recording and reconciliation.

5.14 Notifications

Email templates, bulk queue, shortlist/schedule/offer notices, reminders, logs and retries.

5.15 Reports and audit

Dashboards, applicant counts, scrutiny, GD/PI, merit, seats, offers, accounts, imports and audit history.

6. Import and document integration

The import layer is critical because the external portal remains the source for application submission. Imports must be repeatable, traceable and safe for large datasets.

Channel	Expected operation
Excel / CSV	Upload, select source/cycle, map columns, validate, preview and approve.
REST API	Authorised API fetch/push using documented schemas, credentials and idempotency.
ZIP archive	Upload archive with manifest or stable external application IDs.
Document URLs	Fetch files from trusted, allowlisted URLs through background jobs.
Manual correction	Authorised field correction with before/after values, reason and audit.

Recommended pipeline

Source → import batch → staging records → mapping → validation → duplicate checks → preview/error report → authorised approval → operational records → reconciliation.

Key controls

• Never write raw imports directly into live applicant/application tables.

• Preserve source files, external identifiers, row numbers and import metadata.

• Use source + external application ID + cycle (and relevant application key) for duplicate prevention.

• Keep applicant identity separate from each application; one person may apply to multiple courses/cycles.

• Do not silently overwrite internally verified data; create a field-level update review.

• Process large imports in chunks with background queues, retries and partial-success reports.

• Import application data and documents as separate linked jobs; a missing document must not discard a valid application.

• Validate actual file content, scan for malware, restrict ZIP paths/sizes and prevent unsafe URL fetching.

Policy point to confirm

The institution must decide whether each cycle imports all submitted applicants or only candidates shortlisted by the external portal. The platform should support either through configurable import filters and status mapping.

7. Selection, GD/PI, merit and allocation

Selection policy is configurable by course and admission cycle. Do not hard-code one universal selection model.

Capability	Design requirement
Selection stages	Entrance score, academics, work experience, diversity, written test, GD, PI or other approved stages.
Panel constitution	Multiple panels, chairperson/member eligibility, availability and workload.
Candidate assignment	Assign by course, stage, slot and panel; prevent duplicate/conflicting assignments.
Evaluation criteria	Criteria, maximum marks, weightage, rating scales and mandatory remarks.
Score aggregation	Approved methods; retain individual marks and moderation history.
Merit formula	Version weights/formula; preserve inputs, outputs and formula version.
Tie-break	Configurable, approved tie-break sequence.
Seat matrix	Branch/course/cycle/round capacity and applicable reservation categories.
Allocation	Draft simulation, eligibility/capacity validation, approval and publication.
Waitlist	Policy-aware waitlists and vacancy processing.

Reservation and merit safeguards

- Use only institution-approved reservation and roster rules.
- Open-merit consideration must not be restricted only to unreserved-category applicants.
- Configure vertical/horizontal reservation, PwBD, conversion, supernumerary seats and tie-breaks explicitly.
- Never publish an unapproved draft allocation.
- Record old/new values, reason, actor, time and approval for manual overrides.

Fresh rounds

A fresh round is distinct from normal waitlist movement. It has its own approval, round ID, participating offerings, seat matrix, policy version, eligibility, merit and allocation history. Closed rounds remain read-only unless an authorised reopening is audited.

8. Payment verification and refunds

There is no payment gateway. Applicants pay through institution-approved external channels. This system records and tracks payment particulars and evidence; it does not initiate payments or refunds.

Step	Operation
Payment entry	Accounts records application, branch, course, fee head, amount, date, mode, payer, UTR/challan and evidence.
Submission	Accounts submits entry for review.
Review	Admission Officer/designated authority approves, rejects or returns with remarks.
Verified status	Only approved payment is treated as verified.
Refund request	Record reason, requested/eligible amount, deductions and supporting documents.
Refund approval	Authorised approver approves, rejects or returns.
Refund execution	Actual refund is made outside the system through authorised channels.
Refund recording	Accounts records actual date, amount, mode, reference and evidence.
Reconciliation	Reconcile approved payments, refunds, reversals and outstanding balances.

• Support full and partial refunds and multiple refund transactions.

• Prevent refunds exceeding the refundable balance.

• Never overwrite/delete original payment records; use adjustment, reversal and refund transactions.

• Separate refund request, approval and actual transaction recording where practicable.

• A refund does not automatically reinstate an offer or seat.

9. Database architecture and core schema

Use one central MySQL 8 database with InnoDB, foreign keys, indexed relationship columns, transactions and audit history. Store documents in private object storage; retain metadata and storage keys in the database.

Domain	Principal tables (indicative)
Organisation	institutions, branches, branch_locations, courses, branch_courses, course_offerings
Cycles/rounds	academic_years, admission_cycles, admission_cycle_offerings, admission_rounds
Users/access	users, roles, permissions, user_role_assignments, user_branch_assignments, user_course_assignments
Teams/committees	committees, committee_members, admission_cells, admission_cell_members, accounts_teams, accounts_team_members
Import	external_sources, import_batches, import_files, import_field_mappings, import_records, import_errors, import_update_reviews
Applicants	applicants, applications, external_applications, application_academics, application_entrance_details, application_preferences
Documents	document_types, applicant_documents, document_versions, document_verifications, file_access_logs
Scrutiny	scrutiny_records, eligibility_criteria, eligibility_assessments, eligibility_results, application_status_history
GD/PI	faculty, interview_panels, interview_panel_members, interview_schedules, interview_candidates, evaluation_criteria, interview_evaluations, interview_evaluation_scores
Merit	merit_formulas, merit_components, merit_calculations, merit_component_scores, merit_lists, merit_list_entries, merit_approvals
Allocation	reservation_policies, seat_matrices, seat_allocation_rounds, seat_allocations, waitlists, admission_offers, offer_responses
Admission	admission_confirmations, admission_enrolments

Domain	Principal tables (indicative)
Accounts	fee_heads, application_fee_dues, payment_records, payment_reviews, payment_adjustments, refund_requests, refund_transactions, payment_reconciliations
System/audit	audit_logs, notification_templates, notification_logs, integration_api_logs, system_settings

- Applicant identity is separate from each application.
- Each application links to its course offering, branch, cycle and external application identifier.
- Use unique constraints for external IDs, internal application numbers and round participation.
- Use DECIMAL for marks and money; do not use floating-point for financial amounts.
- Use restrictive deletion rules for approved admissions, merit, payment and audit records.

10. API catalogue

Use a versioned REST API under /api/v1. Apply authentication, server-side permissions, validation, pagination, rate limits and audit logging.

Area	Indicative endpoints
Authentication	POST /auth/login; POST /auth/logout; GET /auth/me
Organisation	GET/POST /branches; GET/POST /courses; POST /branch-courses; GET /course-offerings
Cycles/rounds	GET/POST /admission-cycles; POST /cycles/{id}/rounds; POST /rounds/{id}/close
Users/committees	POST /users; POST /committees; POST /committees/{id}/members; POST /assignments
Imports	POST /imports/excel; POST /imports/csv; POST /imports/api-fetch; POST /imports/documents/zip
Import review	GET /imports/{id}/preview; GET /imports/{id}/errors; POST /imports/{id}/validate; POST /imports/{id}/approve; POST /imports/{id}/execute
Applicants	GET /applicants; GET /applicants/{id}; GET /applications; GET /applications/{id}
Scrutiny	POST /applications/{id}/scrutiny; POST /applications/{id}/eligibility-assessment
GD/PI	POST /panels; POST /schedules; POST /candidate-assignments; POST /evaluations; POST /evaluations/{id}/submit
Merit	POST /merit-calculations/simulate; POST /merit-lists; POST /merit-lists/{id}/approve; POST /merit-lists/{id}/publish
Allocation/offers	POST /allocations/simulate; POST /allocations/{id}/approve; POST /offers; POST /offers/{id}/response
Payments/refunds	POST /applications/{id}/payments; POST /payments/{id}/submit; POST /payments/{id}/approve; POST /refund-requests; POST /refund-requests/{id}/approve; POST /refund-transactions
Reports	GET /reports/dashboard; GET /reports/imports; GET /reports/merit; GET /reports/seat-allocation; GET /reports/accounts

- Use asynchronous job IDs for large imports, exports and bulk notifications.
- Use idempotency keys for external integration and repeatable operations.
- Serve documents through authorised downloads or short-lived signed access.
- Validate workflow state transitions on the server.

11. Technical architecture and capacity

Layer	Recommended design
Frontend	Vue.js 3 responsive web application; role-aware navigation and server-side data grids.
Backend	Laravel modular monolith, REST APIs, policies, services, events and queued jobs.
Database	MySQL 8 InnoDB, indexes, foreign keys, query review and tested backup/restore.
Queue/cache	Redis for background queues, caching and rate-limit support.
File storage	Private scalable object storage; database stores metadata only.
Web serving	Linux, Nginx, PHP-FPM, HTTPS and managed deployment.
Workers	Separate workers for imports, document processing, notifications and report generation.
Monitoring	Application logs, queue monitoring, database metrics, uptime and storage alerts.

Capacity planning targets

Measure	Initial design target
Applicant master	At least 500,000 records

Measure	Initial design target
Application records	Plan for 1,000,000 or more, subject to repeat applications and cycles
Documents	Scalable storage; size from actual file count and average file size
Concurrent users	Initial target of 300, validated by load testing
Bulk imports	Batches up to 100,000 rows using chunked/background processing
Growth	Expansion beyond one million applicants without fundamental redesign

These are design targets, not performance guarantees. Final server sizing depends on concurrency, query complexity, document volume, imports and reporting. Benchmark with realistic datasets before production.

12. Security, privacy and audit

• HTTPS for all application/API traffic; secure password hashing and MFA for privileged accounts.

• Backend role and organisational-scope checks on every protected request.

• Protection against SQL injection, XSS, CSRF, insecure direct object references and mass assignment.

• Private document storage, authorisation, malware scanning and download logging.

• Validate file content, size, archive paths and document URLs; allowlist sources to prevent SSRF.

• Protect and rotate integration credentials; restrict service-account permissions.

• Audit logins, imports, changes, document access, evaluations, approvals, merit publication, payments and refunds.

• Backups with documented restoration tests and restricted backup access.

• Apply the DPDP Act, 2023 and applicable rules/notifications in force at deployment, with institutional privacy, retention and breach-response policies.

• Collect only necessary data; do not collect Aadhaar or other high-risk identifiers without a documented institutional/legal basis.

Indicative service objectives

An interactive request target under two seconds and 99.5% availability may be adopted as initial objectives, subject to hosting design, maintenance windows and load testing. Large imports and exports should run asynchronously and report progress.

13. Reports and dashboards

Report	Key information
Central dashboard	Applicants, applications, branch/course distribution, cycle status and alerts.
Branch/course dashboard	Applications, scrutiny, eligibility, shortlist, GD/PI, merit and admission progress.
Import dashboard	Batch status, processed rows, duplicates, failures, document status and reconciliation.
GD/PI dashboard	Panel workload, schedules, pending evaluations and locked scores.
Merit dashboard	Draft/final merit, score breakdown, approvals, tie-breaks and publication.
Seat dashboard	Sanctioned intake, category matrix, allocations, vacancies, waitlists and offers.
Admission dashboard	Offers issued, accepted, declined, expired, confirmed and pending.
Accounts dashboard	Payments entered, pending review, approved, rejected, refunds and reconciliation.
Audit reports	User activity, overrides, approvals and changes by date, user and module.

Reports should support authorised filters for cycle, round, branch, course, category, status and date range. Large exports should be generated in the background and delivered through secure, time-limited access.

14. Phased development plan

The phases are ordered by dependency. No duration estimates are assigned; effort and schedule should be set after workshops and sample-data assessment.

Phase	Scope and deliverables
1. Foundation	Laravel/Vue project, authentication, RBAC, institution, branches, courses, offerings, cycles, rounds and audit foundation.
2. Import and applicant master	External sources, Excel/CSV/API import, staging, mapping, validation, duplicate controls, applicant/application master.
3. Documents and scrutiny	Document import, private storage, verification, scrutiny checklists, eligibility criteria and decisions.
4. Shortlisting and GD/PI	Entrance-score import, shortlist rules, faculty/panels, scheduling, assignments, evaluation and notifications.
5. Merit engine	Configurable formula, component calculations, tie-breaks, simulation, approval and publication.
6. Reservation and allocation	Seat matrices, approved reservation rules, allocation simulation, approval, offers and waitlists.
7. Admission and accounts	Admission confirmation, manual payment entry/review, refund workflow and reconciliation.
8. Fresh rounds	Fresh-round creation, new imports where permitted, round-specific rules, merit and allocation history.
9. Reporting and go-live	Dashboards, exports, security/load tests, backup/restore, monitoring, training and production deployment.

Critical milestone

Before developing all downstream modules, prove that a representative external dataset can be imported accurately, without duplicate applications, with correct applicant-to-application matching and reliable document association. This is the foundation for the complete system.

15. Testing and acceptance

• Import test: reconcile source row counts, field mappings, duplicates, rejected rows and outcomes.

• Document test: verify matching, private storage, access controls, scanning and failure handling.

• Scale test: load at least 500,000 applicants and representative application/document metadata.

• Concurrency test: validate the initial 300-user target across search, scrutiny, scheduling, evaluation, reports and accounts.

• Access test: confirm users cannot access records outside assigned permissions.

• Workflow test: validate status transitions, approvals, rejections, returns, reopening and closure.

• Merit test: independently verify formulas, normalisation, rounding, tie-breaks and reproducibility.

• Allocation test: validate seat capacity, category rules, waitlist movement and override controls.

• Accounts test: validate payment approval, partial/full refunds, refund caps, evidence and reconciliation.

• Recovery test: restore database and documents from backup and verify integrity.

Go-live acceptance

Production launch requires signed acceptance of core workflows, successful import reconciliation, security review, performance/load-test results, backup restoration evidence, user training and approved operational ownership.

16. Decisions before implementation

Decision	Why it matters
Import population	All applicants or only externally shortlisted applicants; new applicants in later rounds.
External portal	API specifications, export formats, stable IDs, document access and authentication.
Branch/location model	Whether branches and physical campuses are distinct and which courses operate where.
Identity matching	Permanent external applicant ID and duplicate resolution across cycles.
Selection policy	Authority for eligibility, shortlisting, weights, tie-breaks and changes.
Committee governance	Appointment, quorum, approval powers, validity and conflict controls.
Reservation/seat rules	Approved category/roster rules, horizontal reservation, conversions and supernumerary seats.
Payment policy	Fee heads, modes, evidence, approvers, refund deductions and reconciliation.
Notifications	Email/SMS provider, templates, sender identity and approval.
Retention/privacy	Document retention, archive/deletion rules, privacy notice and obligations.
Hosting/operations	Cloud/on-premises, data residency, RPO/RTO, maintenance and support ownership.

Final recommendation

Proceed with one centralised, modular admission platform using Laravel, Vue.js, MySQL 8, Redis and private scalable document storage. Model institution, branches, courses, course offerings, cycles and rounds as connected organisational dimensions. Maintain a common applicant master and separate application records. Provide central administration with fine-grained role- and scope-based access—not separate branch installations or databases.

BASELINE PRODUCT DECISION

One institution-wide platform • One central database • Multiple branches • Multiple courses • Multiple committees and teams • At least 5 lakh applicants • External application portal retained • No payment gateway